Linear Algebra I & Mathematics Tutorial Ib
Nagoya University, Fall 2026

Quiz 3: Rank & functions - Comments

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Contents

1 Comment 1 **2**

2 Comment 2 **2**

3 Comment 3 **2**

4 Comment 4 **2**

5 Comment 5 **3**

1 Comment 1

If you take any any $y \in \mathbb{R}$, then $g(x_1, x_2) = y \implies x_1 - 3x_2 = y \implies x_1 = y + 3x_2$

Now if you take $x_2 = t \in \mathbb{R}$, then $x_1 = y + 3t$ and $x_2 = t$ and this gives infinity may solutions for each $t \in \mathbb{R}$, for one specific $y \in \mathbb{R}$. Hence g is not injective.

2 Comment 2

The same argument as in Comment 1. Because there for any $y \in \mathbb{R}$ in your codomain you get a point in your domain. So, $\text{Im}(g) = \mathbb{R}$.
Hence, g is surjective.

Note: Alternatively, some students gave a really nice argument. They set $x_2 = 0$ and $g(x_1, 0) = x_1$. Now, for any $y \in \mathbb{R}$ we can take $x_1 = y$ as a solution to $g(x_1, 0) = y = x_1$. Hence, easily we get surjectivity.

The students who did this are: **[k]**

3 Comment 3

$f(x_1) = f(x_2) \iff e^{2x_1} - 1 = e^{2x_2} - 1 \iff e^{2(x_1 - x_2)} = 1 \iff 2(x_1 - x_2) = 0 \iff x_1 = x_2$. Hence, f is injective.

4 Comment 4

Below is the correct graph of $f(x) = e^{2x} - 1$. Notice that the graph passes through the origin, since $f(0) = e^0 - 1 = 0$, and that it approaches the horizontal line $y = -1$ as $x \to -\infty$ but never touches it, because $e^{2x} > 0$ for every $x \in \mathbb{R}$.

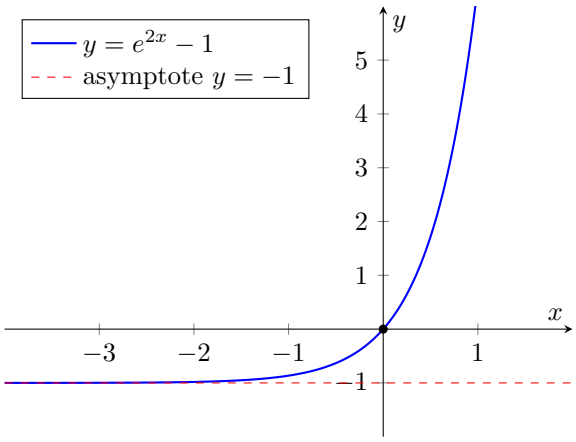


Figure 1: The correct graph of $f(x) = e^{2x} - 1$.

From the graph you can see that $\mathrm{Im}(f) = (-1, \infty)$. So, for example, $y = -2$ has no preimage: $e^{2x} - 1 = -2 \iff e^{2x} = -1$, which is impossible. Hence, $f : \mathbb{R} \to \mathbb{R}$ is not surjective.

Tip. You can use desmos.com to experiment with graphs and learn more. For example, type `y=e^(2x)-1` and try changing the constants to see how the graph moves.

5 Comment 5

In Comment 4 we read the image of $f(x) = e^{2x} - 1$ from the graph. Here is the proof that $\mathrm{Im}(f) = (-1, \infty)$. We show both inclusions.

$\mathrm{Im}(f) \subseteq (-1, \infty)$. Take any $x \in \mathbb{R}$. Since $e^{2x} > 0$, we get $f(x) = e^{2x} - 1 > -1$. So, $f(x) \in (-1, \infty)$.

$(-1, \infty) \subseteq \mathrm{Im}(f)$. Take any $y > -1$. Then $y + 1 > 0$, so $\ln(y + 1)$ is defined. Put $x = \frac{1}{2}\ln(y + 1) \in \mathbb{R}$. Then

$$f(x) = e^{2 \cdot \frac{1}{2} \ln(y+1)} - 1 = e^{\ln(y+1)} - 1 = (y + 1) - 1 = y.$$

So, every $y > -1$ has a preimage.

Hence, $\mathrm{Im}(f) = (-1, \infty)$. But the codomain is $\mathbb{R}$, and $(-1, \infty) \neq \mathbb{R}$. For example, no $y \leq -1$ has a preimage. Hence, $f : \mathbb{R} \to \mathbb{R}$ is not surjective.